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9.3.1 The Power Method

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going to assume that our matrix is diagonalizable. So we have m linearly independent eigenvectors if A is an m by m matrix. And we can write it either like that or like this. Now an algorithm that doesn't compute the eigenvector associated with the largest eigenvalue but that converges to, gets arbitrarily close to, a vector in the correct direction can be given very very simply. Okay, all we do is we take an initial vector v

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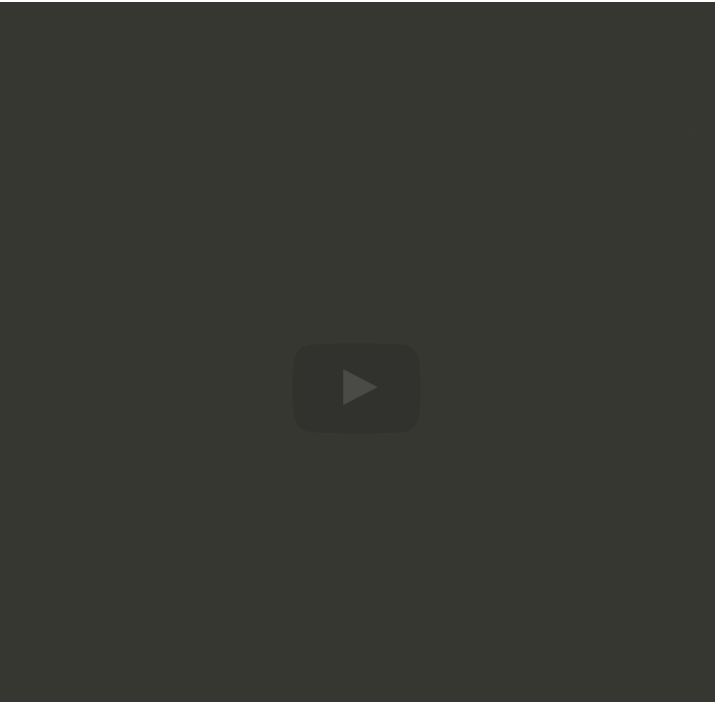
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saying that the magnitude of λ_0 is greater than the magnitude of all of these other eigenvalues. What that means is that as k go to infinity, as we keep doing this iteration, this term is going to end up dominating and all the other terms are going to be inconsequential by comparison.

So how can we make that clearer. Well, we can factor out a λ_0 to the k th power. And then this right here is

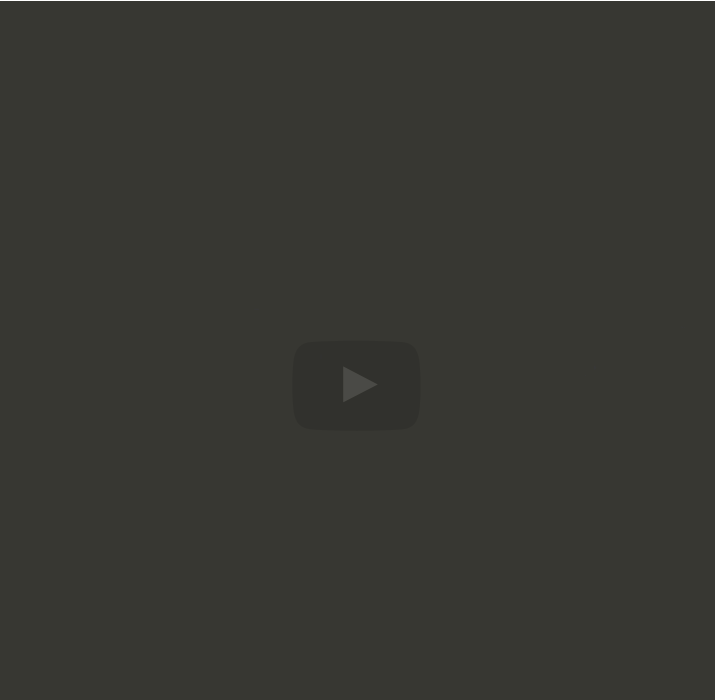
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diagonal elements to the k th power. And then we can again factor out λ_0 to the k th power. In which case this particular element becomes 1 and the rest of these become these ratios. And what we see is that if λ_0 is the largest in magnitude, then these elements all tend to zero. Therefore, this matrix ends up being a matrix that has a standard basis vector in the first column and zeros for the rest. So that

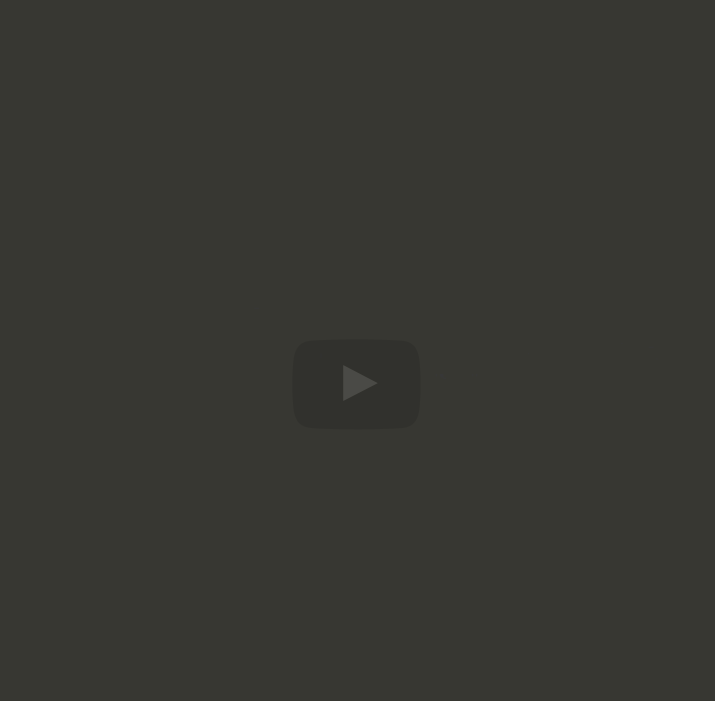
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could form λ_0 times identity minus A . And then we could compute a vector in its null space and we would have our eigenvector. So that's probably cheating, right?

What do we do instead? Well, we recognize that we're only interested in the direction. Alright? So what we could do at every iteration is say, "Okay, let's just make this a temporary vector, and once we've computed it, let's take it and

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Homework 9.3.1.1

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Let x be an eigenvector of A .

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